

ActInf Textbook Group ~ Cohort 2 ~ Meeting 8 (Chapter 4, part 1)

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Greetings, everyone. It is October 21st, 2022. We're in textbook group number 50.1.

No, I just have a slight typo in my OBS. It's cohort two and meeting eight.

We're discussing chapter four in our first discussion of chapter four.

So we're going to, as we did not have any chapter or four specific questions from cohort two, today largely we're going to be reviewing and revising questions that were initially addressed in cohort one, taking a second pass and seeing where we can like add some context and simple or complex or however questions that we can add. So you can be adding in the chat or in the questions tables directly or like just unmuting and addressing them so that we can have high quality and quantity of a large diversity of questions on the material that people are curious about when they're reviewing the text. Before we go to the questions table, does anyone have any general thoughts or comments on chapter four?

Yes, please, Ali.

As we discussed in the previous cohort, this chapter is probably the most technical one in the whole book. So as people go through the formalism here in this chapter, I think it's quite normal to not fully understand every single equation there. So in case there are some difficulties in getting the content of this chapter, I think some of these difficulties, not all of them, would be unpacked in the later chapters.

Okay.

Thank you very much. Totally agreed. Anyone else want to make a general chapter four comment?

Yeah, there's such an interesting rhythm. Chapter one being largely contextualizing and framing. The chapter two, three dialectic, the low road and the high road. Chapter four with a focus on the generative model, and it is a very technical chapter. Chapter five whiplashing over to neurobiology. And less on the fundamental formalisms, more on the neurobiological relationships.

That constitutes the epistemic first half of the book.

The second half of the book has kind of a rhyming pattern, but less focused on epistemic and more on pragmatic. Although, the pragma of epistem. Whereas chapter six is focusing more on the recipe for making the model and less of the contextualizing. Then there's another chapter pair.

Seven and eight, discrete and continuous time. And then chapter nine, which is where we're heading into in just 50 minutes in the first cohort with model based data analysis. And then chapter 10 as sort of like the symmetry of chapter one and the three appendices.

Okay. If anyone has comments, of course, just like raise your hand or speak up or put it in the chat or however.

Okay. I'll un-upvote these so we can upvote them when we've gone through them today.

Okay.

Okay.

Okay.

Well, let's scroll through the chapter before we go to the questions, just to get a view of it.

Okay.

This chapter complements the preceding chapter's conceptual treatment of ACT-INF with a more formal treatment.

It sets out the relationship between free energy, Bayesian inference, form of generative models, and the dynamics obtained from minimizing free energy.

The key focus is on how time is represented.

4.2. Bayesian inference to free energy.

4.3.

Equation 4.1 recalls Bayes' theorem.

Figure 4.1 describes Jensen inequality, which is related to the concavity of the natural log or just the logarithm function in general. And it's a pretty fundamental point. And it speaks to the advantage of using, for example, log probabilities and the different kinds of operations that are possible on logged values.

Here is an application of Jensen's inequality

to derive a free energy value

as bounded,

as bounding.

Rearrangements amongst Bayes' theorem and free energy.

Generative models.

To calculate the free energy, we need three things.

Data, a family of variational distributions, and a generative model comprising a prior and a likelihood.

The section has two types of generative model.

The first is dealing with categorical variables.

The second deals with continuous variables.

Figure 4.2 is helping gain some familiarity with nodes and edges in a Bayesian graph.

So here, in a Bayes' graph, nodes are variables.

Circular nodes are variables.

And edges, which are labeled with a box, reflect their relationships.

So, one can imagine, and we'll return to this in the questions,

these map onto situations where there's a hidden cause and an observation.

You know, temperature in the room, thermometer reading.

Two things that influence one outcome.

One hidden thing that influences two outcomes.

A hierarchical model where something changes something that results in an outcome.

And so on.

And then the boxes are labeled according to the way that we might see that in a more formulaic conditional likelihood expression.

Vertical line means conditional upon.

So, like, box two is the probability distribution of y conditioned upon x .

Because of the directedness of this arrow.

Figure 4.3 is a very central figure in the textbook.

And it sets up the similarities and differences, largely, between the discrete time and the continuous time formulations of active inference generative models in terms of partially observable Markov decision processes.

And for the more Taylor series expansion like approach used for continuous time.

Does anyone want to, like, give a thought or an overview on what is happening in Figure 4.3?

Yes, Ali.

And then anyone else?

Well, the top figure is what happens when we want to model discrete time event.

And the bottom one is more conducive to modeling continuous time phenomena.

So, as we see, they're quite parallel.

And the only difference, or at least the most important difference here is the role that policies play in the discrete time case and the V play in the continuous time case.

The policies denoted with the policies denoted π_i in the top figure.

And the top figure relates to the POMDP approach to modeling discrete time phenomena, as we'll see at length in Chapter 7.

And the bottom one is unpacked in more, I mean, in Chapter 8, I guess.

Excellent.

Again, anyone can raise their hand and go for it, but just to kind of describe what the nodes and edges are here.

We'll be thinking about, like, the temperature in the room is the hidden state and the thermometer is the observation.

So, in the discrete time formulation, this is the temperature in the room through time.

It's the hidden state.

It's in the generative process.

Then, there's an A matrix here labeled as 2, which is the distribution of observations on the thermometer conditioned upon the temperature at that time.

So, the A matrix can be used as a recognition density to do inference on the hidden state given the observation,

or it can be used in a generative capacity given a temperature to emit distributions of thermometer observations.

The 3 corresponds to the B matrix, which is the distribution of the state at the next time conditioned upon the state at the given time and the policy, π_i .

So, $\mathbf{B}$ or $\mathbf{A}$ describes how the temperature is changing through time.

So, this kind of through line of hidden state and then a $\mathbf{B}$ matrix or $\mathbf{A}$ and then a hidden state, this is something changing through time.

And then, π is policy, which is a function of the decision-making of the entity, and it influences how states change through time.

That's the discrete time formulation.

Time point negative 1, 0, and 1, or however.

And often, you see a D or a prior, which is how the first hidden state value gets initialized.

That's the discrete time formulation.

In the continuous time formulation, they're being laid out side by side because the isomorphism of the models is apparent.

They have the same topological structure.

However, there are some really important differences.

Rather than describing how a hidden state changes through time at explicitly demarcated time points, like minute 12, minute 13, minute 14,

what's happening here is that the value of the hidden state at the focal time is still in an $\mathbf{A}$ matrix relationship with observables.

So, the probability of Y given X is exactly analogous to this one of the observations conditioned on the hidden state.

However, taking the place of the $\mathbf{B}$ matrix, $\mathbf{A}$ above here, which describes how hidden states change through time in the discrete time formulation,

is actually an edge that describes the derivative of X conditioned upon X and V ,

which relates to slowly changing causes in the world analogous to policy.

And so, this is actually not doing explicit prediction of future explicit time points,

but rather, it's doing something like a Taylor series expansion,

where as you take more and more derivatives further and further away from a focal point,

you gain better and better approximation through power series expansion,

from a focal point using higher and higher derivatives.

So, here's like an example.

Here's a sine wave,

and the first derivative

just gives you this locally linearizable range.

But, there's a well-defined computational complexity for a Taylor series.

You can say the zeroth order is just zero.

And the first order, just a linear regression.

And then, there's just a quadratic.

And then, there's just a third power relationship.

And so on.

And so, as you get more and more getting into the blue and the purple and so on,

you get better and better fit.

And you extend the range of fit further and further out.

And it always gets strictly better.

This is going to come back in chapter 7 and 8,

which are going to focus more directly on the discrete and continuous time formulations.

But, for here and 4,

they're just being introduced as

different ways that we can have a generative model in ACTIV.

So, now,

Section 4.4 is going to

come to the top of 4.3.

There's some formalisms related to categorical distributions

that are compatible with partially observable Markov decision processes.

Here's an expected free energy G

that describes some different partitionings

on,

as a function of policies,

evaluating policies based upon their impact on the POMDP.

Four key ingredients for the discrete time generative model.

A, B, C, D.

Then, moving to continuous time.

But first, there is

kind of a technical interlude

on Markov blankets.

Previously, there was a box defining the Markov blanket,

and here, it's discussed in the context of message passing.

Yes, Giuseppe, anywhere you want to call attention to.

Yes, thank you, Dan.

I was going back to the figure 4.3.

So, in looking at the parallel between the top and the bottom figure,

so, in the top figure,

π , that is the policy,

is obviously something that is selected by the agent, right?

What about new,

in the, like, I mean,

the Greek letter that is,

that is paralleled to π in the bottom figure,

new, is it,

because you said,
slowly,
slowly varying causes
in the environment,
or something like that.

But,
is that something
emitted,
or selected by the agent,
or is it something,
because,
does anyone have a thought on this?

The distinction is just between discrete and continuous,
is not between, like,
something that,
that it's autonomous,
autonomously selected by the agent,
or something that is,
you know,
happening in the environment,
right?

Ali.

Actually,
in continuous,
in continuous time situation,
the new here can also be
related to
the non-cognitive aspects of the system,
such as the attractors of the system.

So,
that's why
it's described in terms of
the slowly varying causes.

for instance,
in chapter 8,
we will see that,
well,
we have some

static variables,
according to which
the system behaves.

So,
that's not exactly
something
that the system
will choose,
per se.
it's some,
some,
some,
some variables
or some states
towards which
the system
will approach.

So,
that's,
yeah,
that's why
they termed it in battery.

So,
for instance,
when applying to,
to movements,
that can be
some sort of
anatomical constraint
of the movement,
something like that?

Exactly,
yeah.

I see.

Okay,
thank you.

Yeah.

Thank you for

sharing that,

Ali.

It reminds me of the question

that

came later.

Action is part of,

and this is chapter 8,

action is part of

the generative process,

not the generative model.

And so,

the actual

interpretation

of slowly varying causes

and the mapping

of slowly varying causes

to real world phenomena

is

one

interesting

area.

Another

general

point

is that

if we look

to the figures,

we'll see that

the first figure

is the

action perception

loop.

And,

initially,

there's some

usage

of the Bayes graph

in terms of

elaboration

action perception

on the action

perception

loop.

And so on.

But when we get to

chapter 4

and the generative

model,

we don't see

the action perception

loop

represented the same

way.

And so,

it's an interesting

question to ask

where these

variables

map

onto,

for example,

the entity

model.

This isn't

simply an entity

model.

OK.

So,

just completing

our quick

scan before

we go to

the questions.

Message

passing and

mark all

blankets.

This is
definitely an
area that
we could
learn a
lot more
about.

and this
is also
an area
where
software
packages
exist
with the
work of
the
bias
lab
and
their
Forney
package.

OK.

Figure 4-4,
Bayesian
message passing.

Does anyone
want to
describe this
figure?

Just one
thought?

Well,
the overall
thought is,
is this

supposed to
be something
that we're
supposed to
in detail
understand?
Or is it
more like
an
illustrative
representation?

I was
thinking
exactly the
same.

It's like,
here's the
kind of food
you could
make.

Or is
this like,
here's how
you make
this one
dish.

There are
also,
it's not
read from
left to
right,
per se,
at the
level of
the figure.
the
caption

begins
with a
right.
So,
starting with
the left.
The scheme
could be
expanded
hierarchically,
collapsing
over time
steps and
policies for
simplicity.
So,
just looking
at the
left one,
here,
the subscripts
have i
on the
bottom.
Then,
the subscripts
have i
plus one.
But,
you see
the same
squiggle
 π
 s
epsilon.
So,
this is
showing that

within the
context of
a hierarchical
model,
you can
have
recurring
patterns
of
topological
relationships
of a
Bayes
graph to
enable,
for example,
hierarchical
predictive
processing,
active,
Bayesian
action
selection,
et cetera
models.
And then,
that is
collapsed over
time,
steps,
and
policies.
So,
what's
happening
here?
So,
this is

going to
be
unpacking
within
one
layer
on the
right
side.
And,
we see
like,
so,
here,
there's
no
superscript,
but we
can just
take that
to be
like pi
i.
So,
this is
going to
be
this
quartet
being
expanded
out.
So,
here's
o
tilde,
like o
through

time
coming
in.
And,
here's
like,
the
actual
discrete
time
observations
coming
in.
Here's
epsilon
and s
in a
predictive
processing
capacity.
and,
these are
illustrating
with
labeled
edges,
some of
them,
the
relationships
between
these
different
variables.
Squiggle,
in this
case,
that's

funny,
when you
search for
the
squiggle,
it finds
the
other
sigma
as well.

Squiggle
sub
pi t
is
two
natural
logs
being
subtracted.

A natural
log
subtracting
another
natural
log.

So,
what happens
when you
subtract,
what's log
a minus
log b?

It's the
same as
log a
divided
by b.
Log,

open
parenthesis,
a divided
by b .
So,
this is
computing
a ratio
between
the
observation
and
the
preference.
If that
is one,
then
the
incoming
observations
are
exactly
the
preferences,
the
preference
distribution.
if there's
a surplus,
then the
observation
is higher.
If there's
a deficit
below one,
then it
means that
it's

lower.

I hope

that is

the correct

way to

say it.

So,

here's

message

passing

and

relationships

amongst

variables,

and we

see that

kind of

like,

with an

epsilon

intermediating,

we see

the S

relating

to the

O

at a

given

time.

So,

this is

a lot

like

the

A

matrix.

And

here's

the
role
of
policy.
Now,
we're
in the
continuous
time.

4.5.
Ali,
please.

I think
this
figure
can
be
usefully
compared
with
figure

7.12
because
I believe
the
figure
we'll

see in
chapter
7,

7.12,
I guess.
Yes,
that's
it.

Is
the,
I mean,

hierarchy,
the other
interpretation
of this
figure
from
chapter
4.
in
terms
of
its
hierarchical
relationships,
but
from
graphical
probabilistic
modeling
point of
view.
And
the
other
thing
is
that
I
mean,
final
position
sigma,
I don't
think
it's
defined
explicitly
in

the
text,
but
it's
been
defined
as
the
expected
prediction
error
in
the
step-by-step
tutorial
paper.
So,
yes,
that's
also,
I wanted
to
mention
that
definition
as well.
Thank you.
this
step-by-step
tutorial,
which is
also a
four-part
model
stream,
is
very,
very

useful.

In

some

ways,

it is

almost

textbook

length,

whereas

in

chapter

four,

there

was

first

the

general

discussion

of

base

graphs.

like,

again,

one

cause,

one

observation,

one

cause,

two

observations,

two

to

one,

one

to

one

to

one,
and
then
instantly
into the
juxtaposition
between
discrete and
continuous
time.

As we've
looked through
today,
in
contrast,
the approach
taken in
step-by-step
first off
is to
only focus
on the
discrete
time
formulation.

But then
they walk
through a
sequence
of
graphs
and
connect
them to
increasing
levels of
complexity
building

towards
the one
that we
see in
4.3.
It's a
little later
than one
might expect
in the
paper.
So
sorry about
that.
Here it
is.
Starting
with
just
a
simple
graph
that is
exactly
analogous
to the
one on
the top
left.
So they
start with
a static
perception.
Then
in the
step-by-step
they
introduce

this kind
of
E
motif
by showing
a prior
and three
time points
to
changes
observations
mediated
by the
A.
This is
corresponding
to the
music
listening
example
that's
going to
come up
later
in the
textbook.
Then
policy
as
a way
to
influence
state
transition
matrices
is
introduced
on the

top
right
and
free
energy
as
an
objective
function
guided
by
preferences
to do
policy
selection
as
inference
and
then
finally
on the
bottom
right
there
is
some
precision
on
policies
which
is
just
bringing
in
another
variable
beta
and

gamma
which
have
an
inverse
!
relationship
to
each
other
so
that
you
can
talk
about
high
precision
as
low
ambiguity
and
vice
versa
so
it's
a
different
ordering
of
talking
about
generative
models
okay
we're
in the
continuous

time
branch
I'm
just
copying
a link
that
Ali
sent
yes
I
just
sorry
I just
wanted
to
mention
that
the
continuous
time
counter
part
for
that
step
by step
paper
might be
this
one
from
2017
by
Raphael
Bogach
because
in

this
paper
it
almost
exclusively
deals
with
the
continuous
time
active
inference
great
generalized
coordinates
motion
this
is
again
bringing
back
this
idea
of
the
Taylor
series
if
this
black
line
is
the
function
that
has
to
be

approximated

if

you

only

have

one

parameter

to

approximate

it

and

you

want

to

do

it

at

this

line

your

first

is

just

the

actual

value

of

the

function

at

t

equals

zero

then

the

first

derivative

value

at
the
time
x
of
o
and
the
first
derivative
then
adding
in
the
second
derivative
and
so on
so as
you add
more
and more
and more
derivatives
you'll get
better and
better
approximation
to
arbitrary
functions
and
you'll
get
that
with
a
totally

known
form
that
can
be
represented
in
a
vector
as
the
generalized
coordinates
of
motion
and
in
physics
it's
used
in
engineering
it's
used
in
PID
control
this
has
come
up
in
many
live
streams
and
discussions
because

you
can
say
whatever
function
of
approximating
I
want
the
first
five
derivatives
and
then
you
know
that
your
data
structure
is
only
going
to
have
five
coefficients
for
example
and
they've
joked
many
times
like
six
generalized

coordinates

are

sufficient

six

layers

of the

brain

cortex

regions

though

in

some

ways

that's

kind

of

that

makes

it

seem

like

the

six

layers

of

the

cortex

are

like

this

but

actually

the

cortical

column

is

not

like

little
six
floors
where
each
one
is
its
own
predictive
loop
actually
the
layers
are
connected
differently
more
formalisms
in the
continuous
time
setting
with
generalized
coordinates
of
motion
tilde
over a
variable
is
representing
trajectories
one
footnote
is that
this is

still
in the
state
based
base
of
Bayesian
mechanics
or
anyone
else
please
correct
if
this
is
not
the
case
so
the
tilde
here
is
a
summary
notation
helping
us
represent
trajectories
of
states
which
is
moving
us
towards

the
path
formulation
of
Bayesian
mechanics
but
is
not
the
same
as
a
true
path
formulation
hope
that is
fair
to say
the
Laplace
approximation
is
yes
Giuseppe
please
yeah
so
in that
in the
tilde
formalism
now
when we
are talking
about
generalized

coordinates

if we

were talking

about

standard

coordinates

x tilde

would be

x tilde

would be

x at

time

t

comma

x at

time

t

plus

1

x at

time

t

plus

2

etc

now

in the

generalized

coordinates

the

x tilde

would be

x

comma

first derivative

of x

comma

second derivative

of x
is that
correct
yes
so
o tilde
with just
a location
coordinate
would be
either
a list
or a
vector
depending on
whether you
want to
think about
it in
terms of
like a
list
with
comma
or in
terms of
like a
vector
going down
time
yeah
yeah
then
with
generalized
coordinates
it is
a matrix

so
just
it could
be
represented
differently
but
here
let's
just say
that we're
going to
be working
with it
as a
list
like you
said
so
in this
case
columns
correspond
to the
list
above
those
are
the
time
points
rows
going
down
are
the
generalized
coordinates

of
motions
so
for
each
time
point
one
two
three
four
five
we have
a
vector
which
is
describing
the
location
the first
derivative
the second
derivative
and so
on
so
there
would be
six
rows
if we
had six
generalized
coordinates
of
motion
and

this
is
used
and
discussed
in
livestream
26
particularly
and
several
other
places
generalized
coordinates
are
like
they're
kind
of
a
not
like
a
trick
or
anything
to
get
dynamics
into
a
static
representation
like
if you
just
take a

picture
of a
car
you
can't
tell
if
it's
moving
or
not
but
if
you
said
this
is a
generalized
picture
of a
car
it's
at
12
on the
number
line
and
its
velocity
is
10
this
way
and
its
acceleration
it's

slowing
down
and
it's
going
to
the
right
and
the
way
that
it's
slowing
down
is
speeding
up
then
you
can
start
to
say
a
lot
more
about
the
next
steps
of
the
car
exactly
and
that
is

representable

in

a

static

snapshot

another

way

to

looking

at

the

generalized

coordinate

is

to

see

it

as

a

dimension

reduction

technique

because

when

we want

to

describe

as a

very

simple

case

a

trajectory

in

a

cartesian

coordinates

well
we'll
definitely
need
at
least
for
example
in
a
two
dimension
trajectory
will
need
two
dimensions
but
if
we
look
at
it
from
a
generalized
coordinates
perspective
we
can
just
put
it
into
the
Riemann
manifold
and

jettison
one
of the
dimensions
which
encapsulates
that
trajectory
through
a
specific
time
so
one
of
the
dimensions
would
be
reduced
and
then
we
can
just
look
at
the
trajectory
directly
look
at
the
behavior
of
the
trajectory
directly

from
the
consecutive
derivatives
we
take
so
that's
dimension
reduction
technique
as well
nice point
yeah
when you
said
you
said
on the
Riemannian
Riemannian
Riemannian
manifold
you said
can you
explain that
yeah
because in
the
Poincaré
or
Riemannian
manifold
we're
dealing
we're
dealing
with

not
exactly
the
Cartesian
coordinates
but
it's
just
the
state
space
defined
as
these
kind
of
generalized
coordinates
so
it is
already
one
dimension
reduced
coordinate
so to
speak
so
that's
the
whole
point
behind
encompassing
the
state
space
in

that
Riemannian
manifold
not
the
Cartesian
coordinates
because
Cartesian
coordinates
need
some
global
coordinates
to
define
the
states
of
the
space
but
in
generalized
coordinates
it's
more
like
a
kind
of
localized
coordinates
not
global
ones
yes
this is

very
well
I don't
know
I don't
know
much
about
the
Riemannian
or
the
Poincaré
manifold
so
I guess
I'll have
some
readings
to do
to
understand
that
but
yeah
this is
very
deep
and powerful
analytical
insight
one other
way to
think about
generalized
coordinates
as a
dimensionality

reduction
technique
from a
more
computational
applied
perspective
is
if you
want
five
coordinates
of
motion
you
know
exactly
the
size
of
that
data
object
if
you
say
I'm
going
to
consider
20
time
points
and
five
generalized
coordinates
of

motion
you
know
exactly
the
shape
of
that
parameter
or
that
variable
now
whether
that
captures
whether
that
manifold
or
subspace
captures
10
or
90
or
100
percent
of
the
variance
in
the
original
space
is
situational
but

you
will
capture
some
variance
just like
every
principal
component
analysis
will
capture
some
variance
in
the
space
and
whether
it
captures
enough
is
situational
but
you know
you get
some
and
you
know
that
you've
reduced
the
dimensionality
to a
well-defined

amount
which
in
many
cases
is
essential
versus
having
like an
unbounded
number
of
unbounded
!
size
the
motor
reflex
case
has been
especially
studied
from the
continuous
time
perspective
that's
going to
come up
in the
second
half
of the
book
it
comes
up

in
a lot
of
different
other
cases
and in
chapter
five
motor
reflexes
box
4-3
describes
the
Laplace
approximation
I'll just
note again
as
frequently
that
last names
don't really
reduce
uncertainty
about what
something is
talking about
much of
the time
and so
there'd be
ways to
describe
some of
these
things

that
wouldn't
rely as
much
on
specific
last
names
because
there's
like a
many-to-many
mapping
between
the
technique
and
different
last
names
and so
on
but
just
working
with
what
we
have
Laplace
approximation
is
a
quadratic
approximator
that is
an
expansion

around
the
mode
of
a
distribution
so
the
procedure
for
generating
a
Laplacian
approximation
is
you
find
the
mode
of
a
distribution
which
is
the
tallest
point
on
the
distribution
then
an
upside

tip
of
the
parabola

is
going
to
be
the
mode
of
the
distribution
the
width
of
the
parabola
is
going
to
reflect
a
second
parameter
a
variance
parameter
that
captures
as much
as
possible
of
the
original
distribution
so
this
is
kind
of

like
a
good
thing
about
Laplace
approximation
is
that
no
matter
what
shape
the
original
distribution
has
you
know
your
Laplace
approximation
is
only
going
to
need
two
parameters
the
mode
and
the
variance
and
so
in
that

way
it's
a
lot
like
a
Gaussian
approximator
although
the
Gaussian
function
is
different
than
the
quadratic
approximator
they're
similar
because
they
both
have
two
parameters
the
location
of
the
midpoint
and
how
wide
the
function
should
be

stretched
so
for
empirical
distributions
which
might
not
fit
cleanly
to
any
specific
analytical
family
this
can
be
very
helpful
because
again
you
know
that
some
density
of
the
target
distribution
will
be
captured
now
if
you're
dealing

with
like
a
super
tricky
distribution
where
there's
four
humps
all
of
very
similar
height
then
the
Laplace
approximation
is going
to get
tricked
because
it's
only
going
to
capture
like
at
best
25%
of
the
variance
and
it's
going

to
you're
going
to
be
very
much
like
unaware
of
these
other
three
big
peaks
however
four
distributions
that
have
a
central
tendency
broadly
speaking
the
Laplace
approximation
does
extremely
well
some
of
these
subtleties
between
Laplace
approximation

variational
Bayes
and
non-parametric
bootstrap
Bayes
sampling
techniques
are
explored
in
the
SPM
textbook
especially
all
of
this
was
still
inferential
but
not
action
including
and
that
is
very
analogous
to
the
approach
taken
in
the
paper
discussed

in
Livestream
43
which
is
on
predictive
processing
itself
predictive
coding
a
theoretical
experimental
review
in
this
one
80%
or
something
of
the
paper
is
about
hierarchical
predictive
processing
architectures
and
then
they
go
all
right
now
we're

just
going
to
slip
in
action
and
it
turns
out
to
play
very
well
here
we
see
a
figure
that's
kind
of
resonant
with
this
one
we
have
a
discrete
time
formulation
here
we
know
that
because
we

see
like
an
explicit
time
variable
and
like
a
time
minus
one
and
time
plus
one
so
that's
how
one
knows
that
they're
dealing
with
a
discrete
time
formulation
even
if
the
variables
were
called
different
things
in

contrast
we
see
a
hierarchical
architecture
again
but
everything
has been
transposed
into
the
notation
where
we
saw
y
coming
in
whereas
we
saw
o
coming
in
here
and
that
was
what
we
saw
in
4.3
o
coming
in

to
discrete
time
y
coming
in
to
continuous
time
so
this
figure
4.6
is
related
to
the
previous
message
passing
figure
and
then
there's
a
summary
so
chapter
4
as
Ali
mentioned
at the
beginning
is
relatively
technical
there's

a lot
of
techniques
and
notation
and
forays
into
more
general
topics
ranging
from
Jensen's
inequality
at the
beginning
of the
chapter
to
generalized
coordinates
of
motion
message
passing
in the
context
of
Markov
blankets
evidence
lower
bound
and
free
energy
calculations

and
all
of
that
is
being
done
in
this
side
by
side
way
between
discrete
and
continuous
time
and
that
is
broadly
what
chapter
4
describes
all
of
that
is
being
done
because
we're
exploring
generative
models
in

active
inference
okay
in our
final
minutes
what are
some
thoughts
or ideas
that people
have
that we
can come
to
next
week
Yvonne
I'm
fast
now
what does
it mean
to
understand
chapter
4
what are
we
looking
to
get
out
of
chapter
4
I
think

a good
understanding
of
the
discrete
and
continuous
time
distinction
is
like
in my
opinion
the
essence
of
what
is
in
chapter
4
but
I
would
like
to
understand
more
about
the
message
passing
because
I
don't
really
get
it

all
the
way
as
I
would
also
love
to
pick
apart
and
understand
the
larger
paper
on
message
passing
by
first
and
some
colleagues
that
I
don't
remember
yeah
yeah
for
me
the
greatest
challenge
is
that
you

know
even
though
you
you
beginning
to
grasp
the
theoretical
details
the
moving
from
there
to
actually
implementing
something
is
still
quite
difficult
and
you
know
Freestone
says
quotes
a lot
of
time
Feynman
saying
you
can
only
understand

when
you
can
do
it
but
still
it's
that
passage
in
active
inference
is
still
quite
challenging
for
me
even
after
having
read
the
whole
book
and
you
know
and
few
of
the
other
papers
so
this
is

the
even
though
you
know
you
understand
this
this
chapter
four
which
has
a lot
of
the
theory
but
still
going
from
there
to
implementing
a
model
it's
not
easy
I
don't
know
if
there's
something
that
affects
everybody

it
does
affect
me
for
sure
well
certainly
it does
it's
the
fractal
state
of
play
at
this
moment
that's
why
people
aren't
just
like
blasting
out
active
models
because
there's
like
a lot
of
trails
to
blaze
and
also

just
as
always
compare
it
with
a
linear
regression
you
could
read
a
100
500
10
000
page
textbook
on
linear
regression
and
then
you
would
sit
down
in
front
of
your
programming
language
and
you
may
not

specifically
know
how
to
quote
apply
linear
regression
linear
regression
has
had
100
years
of
empirically
focused
development
on
data
input
output
visualization
training
model
adequacy
model
selection
there
are
just
whole
areas
of
applied
statistics
haven't
turned

their
attention
to
this
area
which
is
why
we're
doing
it
so
the
most
ancient
comparator
is
linear
regression

Then
to
get
one
step
closer
to
seeing
what
the
theory
practice
synthesis
looks
like
the
SPM
textbook

documentation

and

literature

help us

see what

it could

look like

and

then

it'll

have

aspects

of

both

of

those

and

some

novelties

all

right

very

well

thank

you

all

for

this

fun

meeting

thank

you

see

you

next

week

thank

you

bye

bye